

Bag of Visual Words(BoVW) for Image Classification:

Notes:

- The used dataset is **UC Merced Land Use Dataset** with **21** different folders, each folder represents a category with **100** .tif images.

To evaluate the performance of the Bag of Visual Words (BoVW) model for image classification, i varied:

- **Dataset Splits:** The dataset was partitioned into: **50-50**, **60-40**, **70-30** and **80-20** where each split was stratified to maintain class balance across all 21 land-use categories.
- **Vocabulary Size (K):** The number of visual words in the dictionary was varied across **21**, **50**, **100**, **150**, **200**, **250** and **300** clusters.
- **Normalization Methods:** Two histogram normalization techniques were tested after visual word quantization: **L1 normalization** and **L2 normalization**.

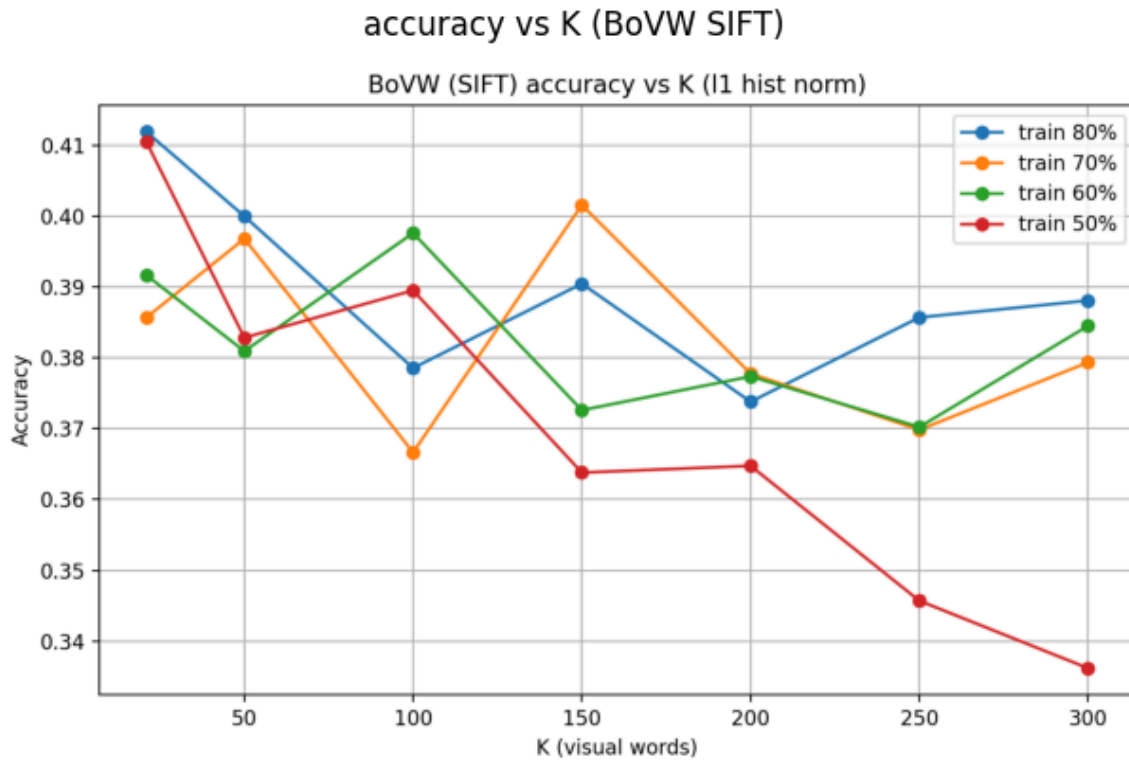
In total, we have 56 different experiments. For each one, SVM classifiers were trained and evaluated. Performance was assessed in terms of **classification accuracy** and **confusion matrices** as we will see.

- In the first experiment we did extract SIFT features from each image on the dataset, compute descriptors and save the result to cache. Then, starting from the second experiment we checked cache and load descriptors directly, we didn't extract SIFT features again.

Obtained results for Normalization = L1:

train\k	21	50	100	150	200	250	300
0.5	0.410476	0.382857	0.389524	0.363810	0.364762	0.345714	0.336190
0.6	0.391667	0.380952	0.397619	0.372619	0.377381	0.370238	0.384524
0.7	0.385714	0.396825	0.366667	0.401587	0.377778	0.369841	0.379365
0.8	0.411905	0.400000	0.378571	0.390476	0.373810	0.385714	0.388095

The best result was 0.411 with k=21 and training set=0.8.

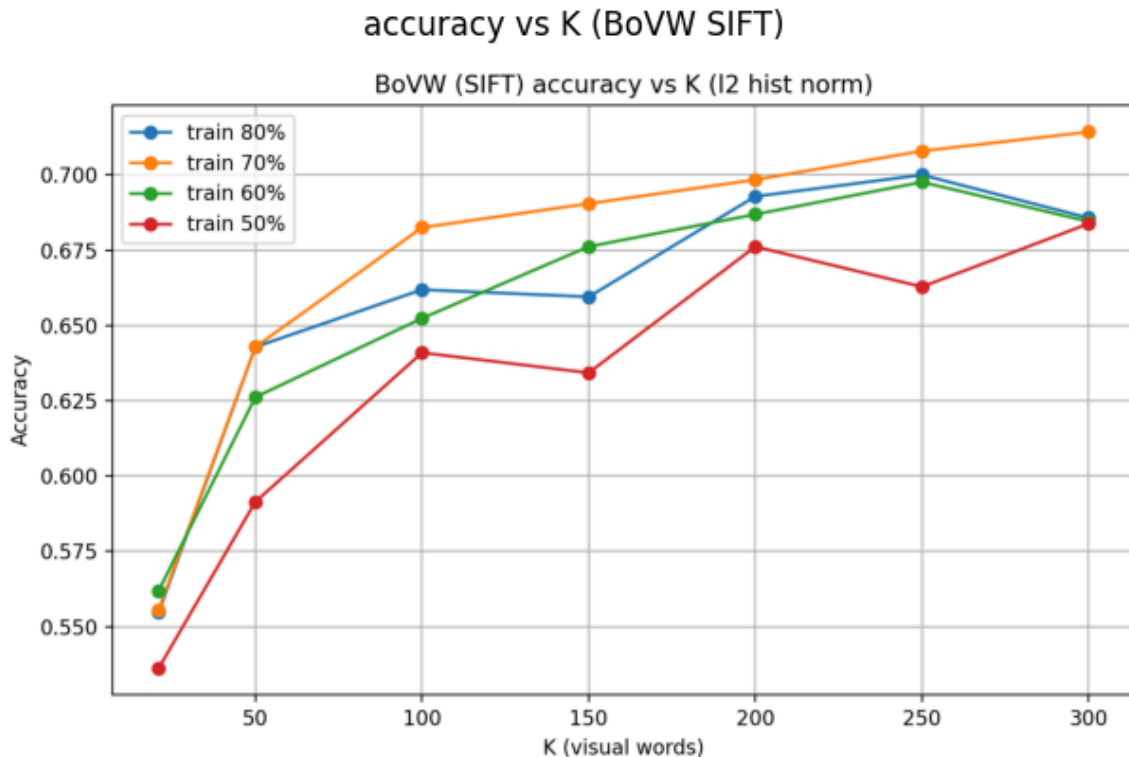


L1 normalization results in weaker and more unstable performance. Across all splits, accuracy remains low and unstable, fluctuating around **0.36-0.41** and increasing the vocabulary size K does not yield consistent improvement. The **L1 normalization** fails to preserve discriminative information in the BoVW histograms because it focuses on relative frequency distribution, it dampens the influence of dominant visual words, which are often critical for class differentiation.

Obtained results for Normalization = L2:

train\k	21	50	100	150	200	250	300
0.5	0.536190	0.591429	0.640952	0.634286	0.676190	0.662857	0.683810
0.6	0.561905	0.626190	0.652381	0.676190	0.686905	0.697619	0.684524
0.7	0.555556	0.642857	0.682540	0.690476	0.698413	0.707937	0.714286
0.8	0.554762	0.642857	0.661905	0.659524	0.692857	0.700000	0.685714

The best result was 0.714 with k=300 and training set=0.7.



The trend is consistent and smooth, indicating stable learning behavior across configurations, it's because of L2 normalization arises from its ability to preserve the geometric structure of the histogram features:

- It ensures that the histogram vectors lie on the unit hypersphere, making their Euclidean distances directly comparable.
- This aligns perfectly with the assumptions of distance-based classifiers (e.g., SVM, k-NN), improving discrimination between image classes.
- In contrast to L1, which only rescales features by total count, L2 balances the contribution of dominant and rare visual words while maintaining global feature magnitude consistency.

So, L2 normalization enhances the separability of BoVW histograms, leading to higher and more stable accuracy across different K values and data splits which makes the model benefits from both a richer vocabulary (higher K) and a consistent normalization that preserves meaningful feature geometry.

Confusion Matrix confusion_70_300_rep0.png

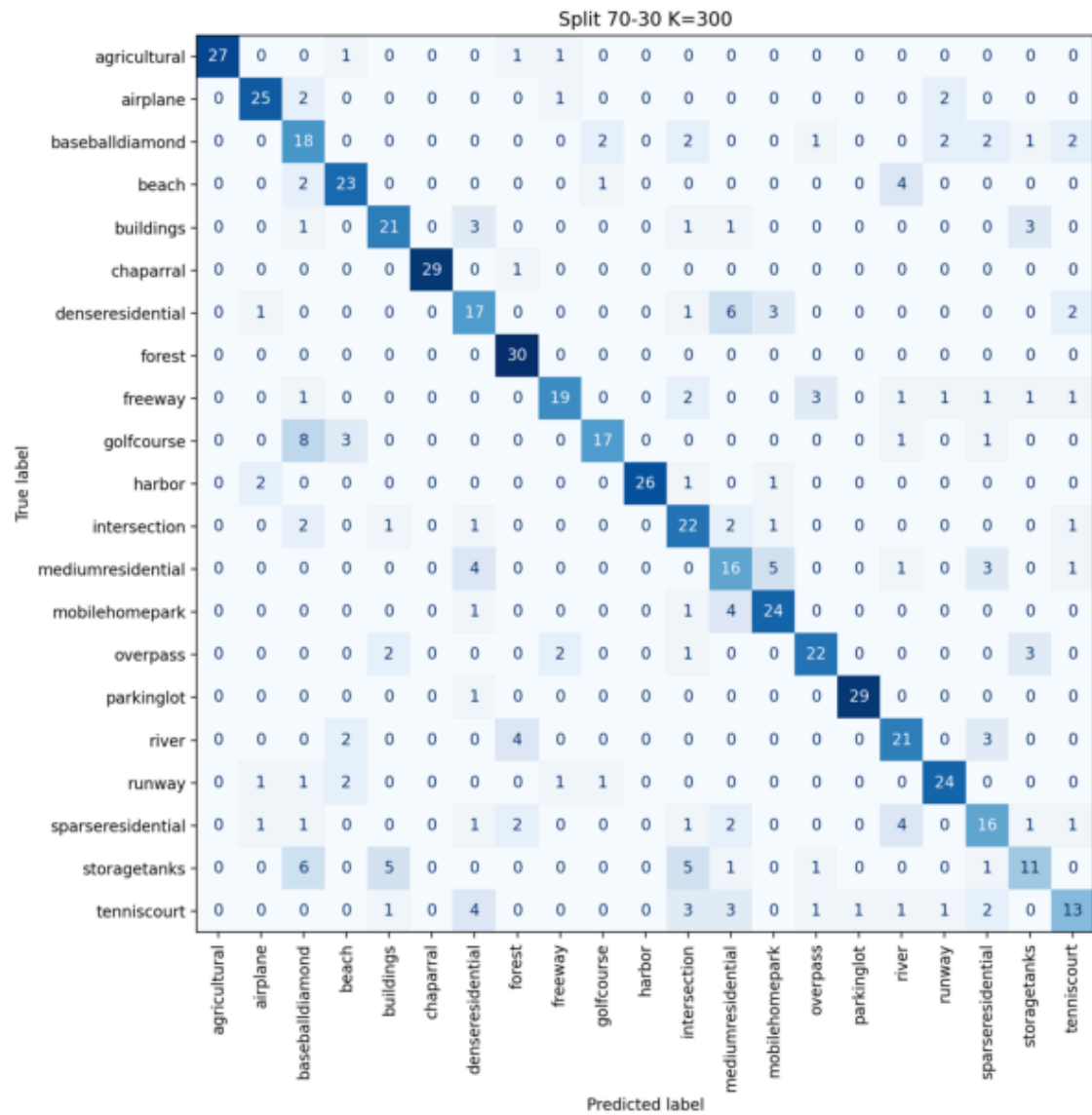
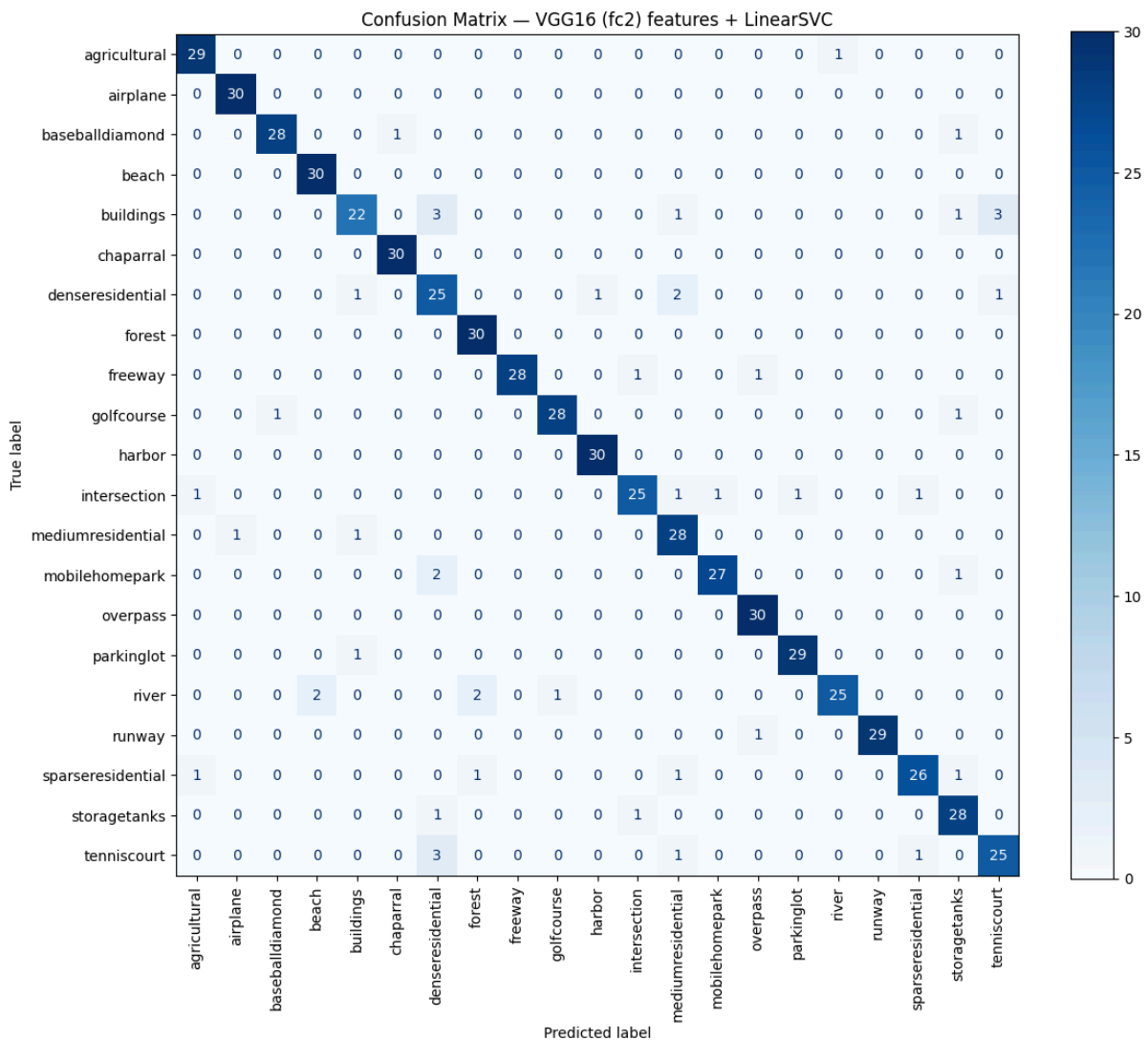


Image Classification Using Deep Features:

- Using VGG16:

In this lab we did load **VGG16** without final classification layer and extract the output from the last fully connected layer, then we resized all the images to **244x244** to match VGG training statistics before processing them in batches to get Numpy array where each row represents an image. We did train a linear SVM wrapped in a pipeline with scaling for normalization.

Results & Discussion:



The obtained accuracy using Deep Features with SVM is **0.9238** where the previous one using BoVW, SIFT and SVM was only **0.714**.

The large improvement in accuracy (from 71.4% → 92.38%) is due to the difference in feature representation power:

- **SIFT(BoVW)** relies on low-level image gradients and keypoints, it cannot capture semantic information or global context where **VGG16 (Deep Features)** uses convolutional layers pre-trained on ImageNet, which encode hierarchical features from textures and edges up to object parts and global structure.
- **BoVW** aggregates local features into histograms and loses spatial relationships where **Deep features** preserve spatial and semantic consistency which is leading to stronger class separability.
- **SIFT/BoVW** is handcrafted, meaning it's human-designed via mathematical rules, gets local descriptors like edges, corners and keypoints, it's limited with the same features for all domains and independent of the data where it uses fixed filters.
- **VGG16** is learned, meaning it learned automatically by the network, gets deep hierarchical representations, it's highly adaptable via training data and trained end to end to minimize classification loss.

● Using ResNet50:

The obtained accuracy was **0.9619**, which was better than using **VGG16 (0.9238)** because of:

1. Architectural Depth and Residual Connections

- ResNet50 is deeper than VGG16 (50 layers vs 16 layers).
- Residual connections in ResNet help **mitigate vanishing gradients**, allowing the network to learn richer, more complex features.
- VGG16 lacks residual connections, which can limit its ability to model very deep feature hierarchies.

2. Feature Representation Quality

- ResNet's bottleneck and skip connections produce **more discriminative embeddings**.
- When used as a feature extractor, ResNet features capture finer details, improving classification accuracy.

Vision Transformer (ViT):

The ViT model achieved the highest classification accuracy among all tested approaches (**97.1%**), slightly surpassing convolution-based architectures such as **ResNet50**.

It can be attributed to several architectural advantages:

1. **Global Context Modeling:**

Unlike CNNs, which capture local spatial patterns via convolutional kernels, ViTs use **self-attention mechanisms** that allow each image patch to attend to every other patch which enables the model to learn **long-range dependencies and holistic representations** of the image.

2. **Data-Driven Feature Learning:**

Whereas SIFT and BoVW rely on handcrafted features that describe local keypoints, ViT learns **task-specific representations** directly from data, adapting feature extraction to the problem domain.

3. **Hierarchical Feature Integration:**

The Transformer encoder aggregates patch embeddings across multiple attention layers, resulting in a **richer, more abstract representation** that improves discriminative power.

4. **Transfer Learning Efficiency:**

The pre-trained ViT (on large-scale datasets such as ImageNet) transfers generalized visual knowledge effectively, even when fine-tuned on smaller or domain-specific datasets.

